

George Baffour Pipim

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Education

- 2021–Present **University of Southern California**, Los Angeles, CA, USA
Ph.D. Physical Chemistry.
Dissertation: First-Principle Modeling of Dynamics and Spectroscopic Signatures in Functional Molecular Systems
Advisor: Professor Anna I. Krylov
- 2021–2024 **University of Southern California**, Los Angeles, CA, USA
M.A. Physical Chemistry
Advisor: Professor Anna I. Krylov
- 2016–2020 **Kwame Nkrumah University of Science and Technology**, Kumasi, Ghana
B.Sc.(First Class Honors) Chemistry, *Graduated top of class*
Thesis: Quantum Mechanical Studies on the Mechanisms of Selected 1,3-Dipolar Cycloaddition Reactions for the Formation of Synthetically- and Pharmaceutically Relevant Precursors
Advisors: Professor Richard Tia & Professor Evans Adei

Professional Appointments

- 2021–2026 **Graduate Research & Teaching Assistant**, University of Southern California, Los Angeles, CA, USA
- 2020–2021 **Teaching Assistant**, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
- 2019–2021 **Research Assistant**, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
- 2016–2017 **Science Tutor**, Jesus Grace Academy, Kumasi, Ghana

Selected Grants, Fellowships, and Awards

- 2024, 2025 **Advancing Science Conference Grant**, National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCCChE)
Competitive travel award supporting participation and research presentation at the NOBCCChE Annual Conference
- 2023–2024 **Dornsife/Graduate School Fellowship**, Graduate School, University of Southern California
Competitive fellowship for excellence in graduate research
- 2021–2026 **Contributor – Novel Toolkit for Harnessing the Power of Exascale Computing for Catalyst Design**, US Department of Energy
PIs: Professor Anna I. Krylov & Professor Shaama Mallikarjun Sharada
- 2021–2022 **Chemistry Graduate Fellowship**, Department of Chemistry, University of Southern California
Competitive fellowship for excellence in graduate research
- 2020 **Overall Outstanding Graduating Student in Chemistry**, College of Science, Kwame Nkrumah University of Science and Technology
In recognition of exceptional academic performance

- 2020 **Outstanding Graduating Student in Chemistry (Male Category)**, *College of Science, Kwame Nkrumah University of Science and Technology*
In recognition of exceptional academic performance
- 2020 **Outstanding Undergraduate Research Award**, *Department of Chemistry, KNUST*
In recognition of exceptional research accomplishments
- 2020–Present **Co-Principal Investigator**, “*Molecular Quantum Chemistry: Reaction Mechanisms and Spectroscopy.*” *Grant Number: CHEM1352, South Africa Centre for High Performance Computing*
Awarded 500,000 core hours (twice every year) on the South Africa’s Centre for High Performance Computing Cluster
- 2019 **College of Science Grant**, *College of Science, Kwame Nkrumah University of Science and Technology*
Merit- and need-based financial award supporting high-achieving students with demonstrated financial need

Research Interests

Electronic structure theory of molecules and materials; density and wavefunction methods for molecular spectroscopy; dynamics of molecules on ground and excited states; computational design of drugs, catalyst and chromophoric systems; and theoretical studies of reactions and energy conversion systems

Research Experience

- 2021–Present **Graduate Research Student**, *University of Southern California, Los Angeles, CA, USA*
Advisor: Professor Anna I. Krylov
- Developed and validated AIMD workflows to study field- and collision-induced fragmentation in EMI-BF₄ propellants; quantified picosecond soft-fragmentation pathways and identified energy thresholds for bond cleavage, informing multiscale thruster models
 - Developed a novel computational toolkit for excited-state modeling of oligo-*p*-phenylenes (OPPs) used for CO₂ photoreduction, integrating TD-DFT/EOM-CCSD with exciton metrics, state-specific emission protocols, and ensemble-based approaches to model exciplex spectroscopic signatures.
 - Modeled Au, Cu, and Ag OLED complexes using X2C Hamiltonians within TD-DFT/EOM-CC frameworks, explicitly treating scalar relativistic effect via two-component formalisms; quantitatively reproduced absorption/emission energies, singlet–triplet gaps, and radiative decay rates, establishing a predictive structure–property framework for next-generation emitters
 - Mapped relaxation pathways of cobalt metalloporphyrins using ab initio molecular dynamics (AIMD) with fewest-switch surface hopping; identified energetically favorable fragmentation channels consistent with ultrafast photodissociation experiments, clarifying competing excited-state relaxation mechanisms
 - Simulated core-level X-ray absorption and circular dichroism (XAS/XCD) spectra of hydrocarbons using EOM-CCSD; assigned spectral features to underlying electronic transitions, providing theoretical insight to support experimental interpretation
 - Built Python tooling to automate job submission, result parsing, and analysis across Q-CHEM/CFOUR/ORCA on Slurm-managed clusters, reducing setup time and human error

2019–2021 **Research Assistant**, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Advisors: Professor Richard Tia and Professor Evans Adei

- Investigated mechanisms of key organic reactions in the synthesis of pharmaceutical intermediates and fine chemicals using DFT, constructing detailed potential energy surfaces to identify transition states, intermediates, and substituent effects governing regio-, stereo-, and chemoselectivity, and deriving insights to rationalize experimental outcomes and guide the development of more efficient, selective synthetic routes.
- Designed and optimized Pt- and Pd-based catalysts for biomass conversion using density functional theory (DFT); investigated reaction mechanisms, active-site energetics, and ligand effects to elucidate structure–activity relationships and guide the development of more efficient catalytic systems for sustainable transformations.
- Performed molecular docking and interaction analysis of small-molecule inhibitors within the active sites of disease-related proteins; characterized key binding interactions and conformational changes to support structure-based drug design and lead optimization.

Peer-Reviewed Scholarly Contributions

*Corresponding author; ‡Student mentee

19. Sampson, D. K.; **Baffour Pipim, G.**; Tingey, N.; Torres, J.; Depew, D.; *Wang, J.; Krylov, A. I., *Collision-Induced Fragmentation of the EMI-BF₄ Propellant in Electrospray Thrusters: Ab Initio Molecular Dynamics Simulations*. J. Chem. Phys. **2026**, **164**, 164308. ([link](#))
18. **Baffour Pipim, G.**; *Sharada, S. M.; *Krylov, A. I., *Photophysical Properties of Functionalized Terphenyls and Implications to Photoredox Catalysis*. Pure Appl. Chem. **2025**, **97**, 1685-1698 ([link](#))
17. **Baffour Pipim, G.**; Sampson, D. K.; Tingey, N.; Torres, J.; Depew, D.; Wang, J.; *Krylov, A. I., *Ab Initio Simulations of Dynamics of EMI-BF₄ Ionic Liquid Propellant Used in Electrospray Thrusters for Nanosatellite Applications*. J. Chem. Phys. **2025**, **163**, 224309 ([link](#))
16. Patra, A.; **Baffour Pipim, G.**; *Krylov, A. I.; *Sharada, S. M., *Performance of Density Functionals for Excited-State Properties of Isolated Chromophores and Exciplexes*. J. Chem. Theory Comput. **2024**, **20** (6), 2520–2537. ([link](#))
15. Sampson, D. K.; **Baffour Pipim, G.**; Tingey, N. N.; Torres, J.; Depew, D.; Wang, J.; *Krylov, A. I., *Exploring Collision-Induced Fragmentation in Porous Electrospray Thrusters*. IEPC-2024-613 (Conference Paper). ([link](#))
14. ‡Atta-Kumi, J.; ***Baffour Pipim, G.**; *Opoku, E., *Tandem [4+2]/Retro[3+2]/[3+2] Cycloaddition Reactions of Fluorinated Oxadiazoles*. J. Phys. Org. Chem. **2023**, **37**(1), e4567. ([link](#))
13. ***Baffour Pipim, G.**; *Opoku, E., *Catalyst-Free (3+2) Cycloaddition Reaction of oxa-, aza-, and thio-bicyclic alkenes with cyclic and acyclic nitrones: A mechanistic study*. Comput. Theor. Chem. **2022**, **1124**, 113790. ([link](#))
12. ‡Fosu, A.; **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Does the Reaction of Nitron Derivatives with Allenates Proceed by an Initial (3+2) Cycloaddition or O-Nucleophilic Addition?* J. Mol. Graph. Model. **2021**, **109**, 108036. ([link](#))
11. ‡Tawiah, A.; **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Exploring the Chemo-, Regio-, and Stereoselectivities of (3+2) Cycloaddition Reactions: A DFT Study*. J. Mol. Model. **2021**, **27**, 287. ([link](#))
10. ‡Ofori, I.; **Baffour Pipim, G.**; *Tia, R.; Adei, E., *A DFT Study of Double (3+2) Cycloaddition Reactions for the Formation of Spirobiisoxazolines*. J. Mol. Graph. Model. **2021**, **109**, 108033. ([link](#))

9. [‡]Atta-Kumi, J.; **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Investigating the Site-, Regio-, and Stereo-Selectivities of the Reactions between Organic Azides and Heteronorbomadiene*. J. Mol. Model. **2021**, **27**, 248. ([link](#))
8. **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Quantum Chemical Investigation of the Formation of Spiroheterocyclic Compounds via (3+2) Cycloaddition*. Tetrahedron **2021**, **94**, 132306. ([link](#))
7. **Baffour Pipim, G.**; *Opoku, E., *Cycloaddition Reactions of Aryl Hetaryl Thioketones and C,N-Disubstituted Nitrilimines*. J. Mol. Model. **2021**, **27**, 84. ([link](#))
6. **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Computational Exploration of 1,3-Dipolar Cycloaddition Reactions*. J. Phys. Org. Chem. **2020**, **34**, e4174. ([link](#))
5. **Baffour Pipim, G.**; *Tia, R.; Adei, E., *(3+2) Cycloaddition Reaction of Benzonorbomadiene and Diazomethane Derivatives: A Theoretical Study*. J. Mol. Graph. Model. **2020**, **101**, 107713. ([link](#))
4. **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Regio-, Stereo-, and Enantioselectivities of Cycloaddition Reactions: A DFT Study*. J. Mol. Graph. Model. **2020**, **100**, 107672. ([link](#))
3. **Baffour Pipim, G.**; Opoku, E.; *Tia, R.; Adei, E. *Peri-, Chemo-, Regio-, and Stereoselectivities of Nitrones with Oxazolones*. J. Mol. Graph. Model. **2020**, **97**, 107542. ([link](#))
2. Opoku, E.; **Baffour Pipim, G.**; *Tia, R.; Adei, E., *Mechanistic Study of Tandem Cycloaddition Reactions for Polyaza Steroids*. J. Heterocycl. Chem. **2020**, **57**, 1748–1758. (Awarded “Top Cited Article” in 2022)([link](#))
1. Opoku, E.; Arhin, G.; **Baffour Pipim, G.**; Adams, A. H.; *Tia, R.; Adei, E., *Site-, Enantio-, and Stereoselectivities of Cycloaddition Reactions: A DFT Study*. Theor. Chem. Acc. **2020**, **139**, 1–15. ([link](#))

Teaching Experience

- 2021–Present **Graduate Teaching Assistant**, University of Southern California, Los Angeles, CA, USA
- CHEM 115B – Advanced General Chemistry II Lab (Honors Class)
 - CHEM 115A – Advanced General Chemistry II Lab (Honors Class)
 - CHEM 108 – General Chemistry Lab (Chemistry Majors)
 - CHEM 105B – General Chemistry II Lab
 - CHEM 322A – Organic Chemistry I Lab (Honors Class)
- 2020–2021 **Teaching Assistant**, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana
- CHEM 473 – Computational Chemistry Class for Seniors (Chemistry Majors)
 - CHEM 369 – Physical Chemistry Lab for Juniors (Chemistry Majors)
 - CHEM 356 – Quantum Chemistry Class for Juniors (Chemistry Majors)
 - CHEM 355 – Physical Chemistry Class for Juniors (Chemistry Majors)
 - CHEM 169 – Physical Chemistry Lab for Freshmen (Chemistry Majors)
 - CHEM 156 – Physical Chemistry Class for Freshmen (Chemistry Majors)
 - CHEM 155 – Physical Chemistry Class for Freshmen (Chemistry Majors)
- 2016–2017 **Science Tutor**, Jesus Grace Academy, Kumasi, Ghana
- Instructed junior high school students in general science and mathematics
 - Organized and supervised science practical sessions to strengthen students’ experimental and analytical skills.
 - Prepared instructional materials and assessments to support effective learning outcomes.
 - Assisted in school information technology operations and digital classroom resources.

Mentoring Experience

Selected Past and Ongoing Mentoring:

- 2020–2021 Mohammed Kassim, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, University of Miami
- 2020–2021 Augustine Obeng, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, State University of New York
- 2020–2021 Michael Appoh, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, Notre Dame University
- 2020–2021 Amos Afugu, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, Wayne State University
- 2020–2021 Emmanuel Forson, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, University of Kansas
- 2020–2021 Michael Eyeson, Mentor — Research, Thesis; (Undergrad, KNUST). *Current*: PhD Student, Purdue University
- 2020–2021 Anthony Tawiah, Mentor — Research, Thesis; first-authored a paper (Undergrad, KNUST). *Current*: Grad Student, Kwame Nkrumah University of Science and Technology
- 2020–2021 Anthony Fosu, Mentor — Research, Thesis; first-authored a paper (Undergrad, KNUST). *Current*: PhD Student, The City University of New York
- 2019–Present Isaac Ofori, Mentor — Research, Thesis (Undergrad, Grad; KNUST). *Current*: PhD Student, Purdue University
- 2019–Present Joshua Atta-Kumi, Mentor — Research, Thesis; first-authored a paper (Undergrad, Grad; KNUST, Nesvard). *Current*: PhD Student, Virginia Polytechnic Institute and State University
- 2019–Present Gabriel Amankwah, Mentor — Research, Thesis (Undergrad, Grad; KNUST, Nesvard). *Current*: PhD Student, Purdue University

Contributed Conference Presentations (*as presenting author*)

9. "Ab Initio Simulation of EMI-BF₄ Ionic Liquid Propellant Used in Electrospray Thrusters", National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE) 52nd Annual Meeting, Hilton Hotel, Atlanta, Georgia, oral presentation, **Sep 2025**
8. "Novel Toolkit for Harnessing the Power of Exaascale Computing for Catalyst Design", US Department of Energy, EFRC-Hub-CMS-CCS Principal Investigators' Meeting, Rockville, Maryland, poster presentation, **Aug 2025**
7. "Computational Toolkit to Design Catalyst for Single Electron Reduction of CO₂", National Conference of the National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE) 51st Annual Meeting, Rosen Shingle Creek, Orlando, Florida, oral presentation, **Sep 2024**
6. "Single Electron Reduction of CO₂ Using Organic Photoredox Catalyst", The 50th John Stauffer Conference, Department of Chemistry, University of Southern California, Los Angeles, CA, poster presentation, **Apr 2024**
5. "Computational Toolkit to Design Catalyst for Single Electron Reduction of CO₂", Modern Wavefunction Methods Summer School, Department of Chemistry, University of Pisa, Pisa, Italy, poster presentation, **Aug 2023**
4. "Single Electron Reduction of CO₂ Using Oligo-p-phenylenes", West Coast Theoretical Chemistry Meeting, Department of Chemistry, University of California, Davis, Sacramento, CA, poster presentation, **May 2023**

3. *"Ab Initio Simulation of EMI-BF₄ Ionic Liquid Propellant Used in Electrospray Thrusters"*, The 49th John Stauffer Conference, Department of Chemistry, University of Southern California, Los Angeles, CA, poster presentation, **Mar 2023**
2. *"Quantum Mechanical Studies on the Mechanisms of Selected 1,3-Dipolar Cycloaddition Reactions for the Formation of Drug Precursors"*, International Virtual Conference on Chemistry and its Application: Schrödinger Day and International Youth Day, Elsevier Research Building Capacity, Department of Chemistry, University of Mauritius, *Aug 2020*
1. *"Quantum Mechanical Studies on the Mechanisms of Selected 1,3-Dipolar Cycloaddition Reactions for the Formation of Drug Precursors"*, International Conference on Materials for Renewable and Sustainable Energy: From Research to Development, Research Workshop for Students, Royal Society of Chemistry – Africa Capacity Building Initiative, Amonoo-Neizer Conference Center, Kumasi, Ghana, poster presentation, **June, 2019**

Community Service, Outreach, and Engagement

- 2023–Present Peer Reviewer, *Scientific Journals & Conferences*: Provide peer reviews of journal manuscripts and evaluated abstracts submitted to chemistry-related conferences - *Theoretical Chemistry Accounts, Journal of Structural Chemistry, Doklady Chemistry Journal*
- 2019–Present Research Student Mentor, *KNUST & USC*: Mentor incoming research students by training them in lab protocols, computational tools, and active projects, and providing ongoing academic and scientific guidance
- Jan 2023 Q-Chem User Workshop, *Winter Virtual School*: Mentored participants during live computational exercises and provided real-time technical support for Q-Chem simulations
- Fall 2024 USC Chemistry Recruitment Guide: Assisted with hosting and guiding prospective students and supported departmental outreach activities
- 2019–2022 Program Officer, *TopGates Africa*: Assisted in planning and coordinating programs aimed at improving education in rural communities
- 2017–2020 Peer Counsellor, *Science Students' Association, KNUST*: Provided academic and personal guidance to first-year students
- 2019–2020 Chairman of Academic Committee, *Ghana Students' Chemical Society, KNUST*: Organized academic support and mentorship programs for first-year chemistry students
- 2019–Present Research Fellow, *Nesvard Institute of Molecular Sciences, Accra, Ghana*
- Mentor students from Africa, grooming their confidence and identity, and helping them develop critical skills for success in graduate school and professional careers.
 - Mentor African-based high school students, college students, and recent university graduates in general scientific research, with a focus on computational chemistry.
 - Conduct computational chemistry research with mentees, focusing on molecular spectroscopy and reaction mechanisms using density functional theory (DFT) and wavefunction-based methods.

Professional Affiliations

Member of the following associations/societies: American Chemical Society, Ghana Chemical Society, National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCCChE), and International Younger Chemists Network.

References

- **Professor Anna I. Krylov**
Department of Chemistry, University of Southern California, Los Angeles, CA (Email: krylov@usc.edu)
PhD Advisor

- **Professor Aiichiro Nakano**

Viterbi School of Engineering, University of Southern California, Los Angeles, CA (Email: anakano@usc.edu)

PhD Advisory Committee Member

- **Professor Shaama Mallikarjun Sharada**

Viterbi School of Engineering, University of Southern California, Los Angeles, CA (Email: ssharada@usc.edu)

PhD Research Collaborator